

MATH 708. SPRING 2026. HOMEWORK 5. DUE ON MARCH 13.

- Unless otherwise stated, the exercises are from Voisin's textbook.
- The homework problems may be submitted orally (during assigned office hours), in written/typed form, or as a combination of the two.
- You may submit any subset of problems totaling up to 100 points.
- If you submit more than 100 points worth, I will grade only the first 100 points you designate.
- Do not write anything on the cover sheet, as it will be used for grading. Bring it to office hours when you submit problems orally, and staple it to your written/typed solutions.

Problem 1. (10 points) Recall that the connected sum $M \sharp N$ of two smooth n -manifolds M and N is defined as follows. Choose smooth embeddings of the unit ball $\mathbb{D} = \{v \in \mathbb{R}^n : \|v\| < 1\}$ into M and into N . Remove the closed balls of radius $1/2$ from these charts and glue the punctured balls using the inversion map $\phi(v) = \frac{1}{2} \|v\|^{-2} v$, so that ϕ identifies the annulus $\{1/2 < \|v\| < 1\}$ with itself while interchanging the two boundary spheres. If M and N are oriented, then $M \sharp N$ inherits an orientation provided the chosen identifications of U and V with $\mathbb{D}$ induce opposite orientations on it (the inversion map ϕ is orientation-reversing).

Now let X be a complex manifold of complex dimension n , and let $\text{Bl}_p(X)$ be its blow-up at a point $p \in X$. Using explicit local equations for the blow-up, prove that $\text{Bl}_p(X)$ is diffeomorphic to $X \sharp \overline{\mathbb{CP}}^n$, where $\overline{\mathbb{CP}}^n$ denotes $\mathbb{CP}^n$ with the opposite orientation to the canonical one.

Problem 2. (10 points) Let X be a complex manifold, and let $\mathcal{O}_X$ denote the sheaf of holomorphic functions on X . Show that the category of holomorphic vector bundles on X is equivalent to the category of locally free $\mathcal{O}_X$ -modules of finite rank, by establishing the following steps.

- If E is a holomorphic vector bundle of rank r on X , show that its sheaf of local sections $\mathcal{E}$ is a locally free $\mathcal{O}_X$ -module of rank r .
- If $E, F \rightarrow X$ are holomorphic vector bundles with sheaves of sections $\mathcal{E}, \mathcal{F}$, construct a natural map

$$\text{Hom}_{\text{VB}(X)}(E, F) \longrightarrow \text{Hom}_{\mathcal{O}_X}(\mathcal{E}, \mathcal{F})$$

and show that it is injective.

- Prove that the map in (b) is surjective.
- Show that every locally free $\mathcal{O}_X$ -module $\mathcal{G}$ of finite rank is isomorphic to the sheaf of holomorphic local sections of some holomorphic vector bundle $G \rightarrow X$.

Conclude that the category of holomorphic vector bundles on X is equivalent to the category of locally free $\mathcal{O}_X$ -modules of finite rank.

Problem 3. (10 points) Let X be a complex manifold. A sheaf $\mathcal{F}$ of $\mathcal{O}_X$ -modules is said to be *generated by global sections* if, for every point $x \in X$,

there exist global sections $s_i \in \Gamma(X, \mathcal{F})$, $i \in I$, such that the stalk $\mathcal{F}_x$ is generated, as an $\mathcal{O}_{X,x}$ -module, by the germs $(s_i)_x$.

Show that $\mathcal{F}$ is generated by global sections if and only if $\mathcal{F}$ is isomorphic to a quotient of a free sheaf of $\mathcal{O}_X$ -modules.

Problem 4. (10 points) Let X be a topological space. Show that taking global sections, $\mathcal{F} \mapsto \Gamma(X, \mathcal{F})$, defines a functor from the category of sheaves of abelian groups on X to the category of abelian groups. Give an example showing that this functor is not exact; that is, it does not necessarily take a short exact sequence of sheaves to a short exact sequence of abelian groups.

Problem 5. (10 points) Given a morphism of sheaves $\phi : \mathcal{F} \rightarrow \mathcal{G}$:

- (a) Show that the kernel presheaf $\ker(\phi)$ is a sheaf.
- (b) Show that ϕ is injective if and only if $\ker(\phi) = 0$.
- (c) Show that the image presheaf need not be a sheaf. Let $\text{im}(\phi)$ denote its sheafification.
- (d) Show that ϕ is surjective if and only if $\text{im}(\phi) = \mathcal{G}$.
- (e) Show that $\text{im}\phi \simeq \mathcal{F}/\ker\phi$.

Problem 6. (10 points) Let $\mathcal{F}$ be a presheaf of abelian groups on a topological space X . For an open set $U \subset X$, define

$$\widehat{\mathcal{F}}(U) = \left\{ s = (s(x))_{x \in U} \in \prod_{x \in U} \mathcal{F}_x \mid \begin{array}{l} \forall x \in U \exists \text{ an open set } V \subset U \text{ with} \\ x \in V, \exists \text{ a section } t \in \mathcal{F}(V), \\ \text{and } \forall y \in V \text{ we have } s(y) = t_y, \\ \text{where } t_y \in \mathcal{F}_y \text{ is the germ of } t \text{ at } y \end{array} \right\}.$$

Show that $\widehat{\mathcal{F}}$ is a sheaf on X , and that the natural morphism of presheaves $\mathcal{F} \rightarrow \widehat{\mathcal{F}}$ exhibits $\widehat{\mathcal{F}}$ as the sheafification of $\mathcal{F}$.

Problem 7. (5 points) Let $\mathcal{F}$ be a sheaf of abelian groups on X , and let $s \in \mathcal{F}(U)$ be a section over an open set $U \subset X$. The *support* of s is

$$\text{Supp}(s) = \{ p \in U \mid s_p \neq 0 \text{ in the stalk } \mathcal{F}_p \}.$$

Show that $\text{Supp}(s)$ is closed in U .

Problem 8. (10 points) Let $\mathcal{F}$ and $\mathcal{G}$ be sheaves of abelian groups on X .

- (a) Show that the assignment

$$U \mapsto \text{Hom}(\mathcal{F}|_U, \mathcal{G}|_U)$$

defines a sheaf on X , called the sheaf $\mathcal{H}om(\mathcal{F}, \mathcal{G})$ of local morphisms.

- (b) Show that $\text{Hom}(\mathcal{F}, \cdot)$ (respectively, $\mathcal{H}om(\mathcal{F}, \cdot)$) is a left-exact functor from the category of sheaves of abelian groups on X to the category of abelian groups (respectively, to the category of sheaves on X).

Problem 9. (10 points) Let $\{\mathcal{F}_i\}$ be a direct system of sheaves of abelian groups on X with transition morphisms $\mathcal{F}_i \rightarrow \mathcal{F}_j$.

- (a) Show that the assignment $U \mapsto \varinjlim_i \mathcal{F}_i(U)$ defines a presheaf on X .
- (b) Show that the sheafification of this presheaf is the direct limit of the system $\{\mathcal{F}_i\}$ in the category of sheaves.

Problem 10. (15 points) Let $f : X \rightarrow Y$ be a continuous map of topological spaces.

- (a) Let $\mathcal{F}$ be a sheaf on X . Define the *pushforward sheaf* $f_*\mathcal{F}$ on Y by

$$(f_*\mathcal{F})(V) = \mathcal{F}(f^{-1}(V)) \quad \text{for every open set } V \subset Y.$$

Show that $f_*\mathcal{F}$ is a sheaf on Y , and that $\mathcal{F} \mapsto f_*\mathcal{F}$ defines a functor from sheaves on X to sheaves on Y .

- (b) Let $\mathcal{G}$ be a sheaf on Y . Define a presheaf on X by

$$U \longmapsto \varinjlim_{f(U) \subset V} \mathcal{G}(V),$$

where the direct limit is taken over all open subsets $V \subset Y$ containing $f(U)$. Let $f^{-1}(\mathcal{G})$ be the sheafification of this presheaf. Show that $\mathcal{G} \mapsto f^{-1}(\mathcal{G})$ defines a functor from sheaves on Y to sheaves on X .

- (c) Prove that f^{-1} is a left adjoint functor of f_* ; that is, for any sheaf $\mathcal{F}$ on X and any sheaf $\mathcal{G}$ on Y , there is a natural bijection

$$\mathrm{Hom}_X(f^{-1}\mathcal{G}, \mathcal{F}) \cong \mathrm{Hom}_Y(\mathcal{G}, f_*\mathcal{F}).$$

Problem 11. (10 points) Let $\mathcal{F}$ and $\mathcal{G}$ be sheaves of $\mathcal{O}_X$ -modules on a complex manifold X . Show that the assignment

$$U \longmapsto \mathcal{F}(U) \otimes_{\mathcal{O}_X(U)} \mathcal{G}(U)$$

defines a presheaf on X , but that it is not necessarily a sheaf. Its sheafification is denoted by $\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G}$. Compute its stalks.

Problem 12. (15 points) Page 115, Exercise 1.

Problem 13. (5 points for each of (a), (b), (c), (d)) Page 115, Exercise 2.